

# ATMOSYNC Project Report

IoT Data Generation, Kafka Streaming, Snowflake Data Analysis & Power BI Dashboard

## 1. Project Overview

ATMOSYNC is an IoT-based data pipeline and analytics project designed to simulate container sensor data, stream the data through Apache Kafka, store and analyze the data in Snowflake, and present meaningful business and operational insights through Microsoft Power BI.

## 2. Project Objectives

- Generate realistic container sensor data using a Python simulator.
- Stream the generated telemetry data using Apache Kafka.
- Verify successful data transmission using a Kafka consumer.
- Store structured datasets in Snowflake for analysis.
- Perform data validation, cleaning, aggregation, joins, and analytical SQL queries.
- Build an executive Power BI dashboard for monitoring and decision-making.

## 3. End-to-End Workflow

Python Simulator → Kafka Producer → Container Telemetry Topic → Kafka Consumer → Snowflake → SQL Analysis → Power BI Dashboard

### Workflow Description

| Stage  | Activity           | Description                                                                                                                          |
|--------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Step 1 | Data Generation    | A Python-based simulator generates container IoT sensor data such as temperature, humidity, and vibration.                           |
| Step 2 | Kafka Streaming    | A Python Kafka Producer sends the generated telemetry data to the container_telemetry Kafka topic.                                   |
| Step 3 | Data Verification  | A Kafka Consumer is used to verify that the streamed data is being received successfully.                                            |
| Step 4 | Snowflake Storage  | The data is stored in the Snowflake environment using the ATMOSYNC_Proj_2_DB database, RAW schema.                                   |
| Step 5 | SQL Analysis       | SQL queries are used for validation, duplicate checks, aggregations, joins, sensor analysis, shipment details, and commodity prices. |
| Step 6 | Power BI Dashboard | The analyzed data is connected to Power BI to create an executive dashboard containing KPIs, charts, and maps.                       |

## 4. Snowflake Data Layer

The Snowflake environment uses the database **ATMOSYNC\_Proj\_2\_DB** and a RAW schema for storing the source datasets. The four CSV datasets are maintained in separate tables because they contain different columns and represent different business entities.

| CSV Dataset       | Snowflake Table       | Purpose                                  |
|-------------------|-----------------------|------------------------------------------|
| IoT Sensor Data   | RAW.IOT_SENSOR_DATA   | Container sensor telemetry               |
| Container Details | RAW.CONTAINER_DETAILS | Shipment, route and quantity information |
| Commodity Price   | RAW.COMMODITY_PRICE   | Commodity and market price information   |

| CSV Dataset    | Snowflake Table    | Purpose                           |
|----------------|--------------------|-----------------------------------|
| Market Details | RAW.MARKET_DETAILS | Market and geographic information |

## 5. SQL Data Analysis

The SQL analysis phase includes the following activities:

- Data validation and record-count checks.
- Duplicate detection and duplicate handling using GROUP BY, HAVING, and ROW\_NUMBER().
- Average, minimum, and maximum sensor readings.
- Container-wise and product-wise temperature, humidity, and vibration analysis.
- Shipment, source, destination, route, and quantity analysis.
- Commodity price analysis by commodity and market.
- Market distribution and geographic analysis.
- JOIN operations between IoT Sensor Data and Container Details.
- Identification of potentially high-risk sensor readings using threshold-based conditions.

## 6. Power BI Executive Dashboard

The final dashboard is designed to provide a single-screen executive view of logistics and IoT performance.

### Key Performance Indicators (KPIs)

| KPI                     | Source Table      | Purpose                         |
|-------------------------|-------------------|---------------------------------|
| Total Containers        | CONTAINER_DETAILS | Total unique containers         |
| Total Shipments         | CONTAINER_DETAILS | Total shipment records          |
| Total Quantity          | CONTAINER_DETAILS | Total transported quantity      |
| Average Temperature     | IOT_SENSOR_DATA   | Overall average temperature     |
| Average Humidity        | IOT_SENSOR_DATA   | Overall average humidity        |
| Alert Readings          | IOT_SENSOR_DATA   | High-risk sensor readings       |
| Average Commodity Price | COMMODITY_PRICE   | Overall average commodity price |

### Recommended Dashboard Visuals

- Shipment by Source and Destination – route comparison.
- Quantity by Product – transported quantity comparison.
- Temperature and Humidity Trend – sensor trend over time.
- Commodity Price by Market – market price comparison.
- Shipments by Source City – share of total shipments.
- Quantity by Product – share of total transported quantity.
- Slicers: Product ID, Container ID, Source, Destination, Date, Commodity, Market, and State where relevant.

## 7. Key Outcomes

- Established a complete data flow from simulated IoT data to business intelligence reporting.
- Successfully streamed and verified telemetry data through Kafka.
- Structured multiple CSV datasets in Snowflake for centralized analysis.
- Performed SQL-based data quality checks and analytical processing.
- Combined sensor and logistics information using SQL JOIN operations.
- Created an executive-level Power BI dashboard for operational monitoring and insight generation.

## 8. Conclusion

The ATMOSYNC project demonstrates a practical IoT analytics pipeline in which simulated sensor data is generated with Python, streamed through Kafka, stored and analyzed in Snowflake, and transformed into actionable insights through SQL and Power BI. The solution provides a foundation for monitoring container conditions, shipment activity, commodity prices, and market information in a unified analytics environment.